

High Aspect Ratio DRIE Development — Project Passdown

Bosch-process Deep Reactive Ion Etching optimization using Design of Experiments (DOE), Washington Nanofabrication Facility (WNF), University of Washington

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Note for the AI assistant reading this file: this is a project passdown for a silicon DRIE (Deep Reactive Ion Etching) recipe-development project, compiled from the outgoing student's proposal, poster, slide decks, final report draft, lab data spreadsheets, reference literature, and a series of clarifying conversations with the outgoing student (Mick). It's meant to give incoming students — and any AI helping them — enough shared context to answer questions about the process, the data, and what to do next.

Two things worth knowing about how this document was built: (1) several numbers in the source spreadsheets are ambiguous or under-documented — the spreadsheet itself needs cleanup, and that's flagged explicitly below rather than papered over; (2) raw SEM (electron microscope) images referenced throughout are **not included in this file** — they live as ZIP archives in a companion GitHub repository (Section 12) that an AI assistant can fetch/unzip and search by wafer ID to answer image-specific questions that come up while reviewing the Excel files. Where this document states a number with a caveat (e.g., "reported ~60:1, but..."), treat that caveat as load-bearing, not decorative — don't round it off to a clean answer when someone asks.

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1. Project overview & goal

This project develops a Bosch-process Deep Reactive Ion Etching (DRIE) recipe on the WNF's SPTS Silicon DRIE tool capable of etching silicon trenches and holes with very high **aspect ratio** — the ratio of feature depth to feature width. The target is an aspect ratio between **50:1 and 100:1**, well beyond what earlier WNF projects had achieved (previous oxide-hard-mask work topped out around 25:1).

The approach is iterative: build a repeatable test-wafer fabrication process, etch on the DRIE tool, measure the results (etch rate, mask selectivity, sidewall profile, aspect ratio), feed the data into **JMP statistical DOE software** to choose new parameter combinations, and repeat. The eventual deliverable is a validated high-aspect-ratio Bosch recipe that the WNF can offer to its industrial and academic users (e.g., for MEMS devices such as accelerometers, gyroscopes, and microfluidic systems).

2. Background concepts

Reactive Ion Etching (RIE) and the Bosch process

RIE uses plasma and an applied electric field to direct free ions at a silicon wafer, driving chemical reactions that remove exposed material. A plain RIE etch tends to lose depth and verticality as it gets deeper, because it becomes harder to direct ions precisely into a narrow, deep feature.

The **Bosch process** addresses this by cycling through three sub-steps in rapid succession, repeated many times ("loops"):

- **Passivation (deposition) step** — a thin protective polymer film (from C_4F_8) is deposited over the whole exposed surface, including the sidewalls.
- **Depassivation step** — a short, more directional step that selectively strips the passivation film off the *bottom* of the trench (where ion bombardment is most vertical) while leaving the sidewall passivation largely intact.
- **Etch step** — an SF_6 -based plasma etches downward into the now-exposed silicon at the trench bottom; the sidewalls stay protected because their passivation wasn't stripped.

Correction from an earlier draft of this document: the Bosch cycle here is a three-step passivation/depassivation/etch loop, not a simple two-step deposit/etch alternation — this matches how the actual recipe files are structured (see Section 5) and how the DOE varies "Passivation Pressure," "Depassivation Pressure," "Platen Power," and "Etch Pressure/Time" as separate parameters.

Aspect ratio and Aspect-Ratio-Dependent Etching (ARDE)

Aspect ratio = etch depth ÷ feature width. As aspect ratio increases, it becomes harder for ions and neutral radicals to reach the bottom of the trench, so etch rate slows and sidewall profile can degrade — this general phenomenon is called aspect-ratio-dependent etching (ARDE). One lever for improving aspect ratio is **lowering etch pressure**: at higher pressure, ions scatter off-axis before reaching the trench bottom (leading to bowed or narrowing sidewalls); at lower pressure, ion trajectories stay straighter. In this project's DOE data, pressure changes alone had a smaller effect than expected — see Section 7. Also see Section 8 for why "aspect ratio" as reported in this project's data is not a single clean number.

3. Wafers, materials & tools

Test wafer stack

- 1 mm thick, 100 mm diameter silicon wafers used for all work to date (batch 1: A-series; batch 3: B-series). **Planned change:** switch to **500 μ m thick** wafers for the next mask/DOE round — cheaper, and tools handle them better. 1 mm-thick wafers are only

necessary if etching deeper than 500 μm , which shouldn't be needed once the mask moves to smaller features. See the feature-size caution in the callout below.

- SiO_2 deposited by PECVD (PECVD2 tool) — this oxide is the **hard mask** for the deep silicon etch. **Current direction: revert to 4 μm oxide** (Mick's recommendation, final call left to Caitlynn/Samyama) — batch 3 had planned a thicker 6 μm oxide, but the lithography ("photo") burden of coating and patterning correspondingly thicker resist isn't worth it. There's no clear benefit to moving off 4 μm right now.
- AZ10XT photoresist spin-coated on top of the oxide — **7 μm** , matching the 4 μm oxide target above. (A 9 μm , and later a considered 10 μm , resist thickness was planned specifically to support the now-deprioritized 6 μm oxide plan — that revision is effectively shelved along with it.)
- Patterned with the WNF "HAR DRIE" / "Uniformity" etch mask: trenches ranging from 5 μm to 500 μm wide. This was used for all work to date. A new mask (see Section 11) with smaller, more numerous features is planned, most likely via maskless lithography rather than a physical photomask.

Open investigation: maximum safe feature size for the new mask

OPEN — ASSIGN TO STUDENTS

Moving to smaller trenches/pillars and a thinner (500 μm) wafer raises a real risk: if a feature — especially a narrow silicon *pillar* between trenches — collapses or is otherwise lost during etching, the newly-exposed area etches much faster than intended, and can etch all the way through the (thinner) wafer. A full breakthrough will halt/abort the recipe. Before finalizing the new mask's minimum feature size, someone should **parse the existing SEM images and etch-rate data to estimate a safe minimum feature size** (i.e., how narrow can a pillar or trench wall get before it's at meaningful collapse risk). A good starting point: image `A14_C_01_e.jpg` in the SEM archive (Section 9) shows one wafer's etch depth across many trench widths side-by-side in a single annotated cross-section — useful raw material for this analysis, though a full answer will likely require looking across several wafers.

Gases and consumables

SiH_4 , N_2 , O_2 , SF_6 , C_4F_8 , Ar; AZ10XT photoresist; AZ400K developer; HMDS vapor prime; EKC830 photoresist stripper.

Tools used (WNF)

Tool	Role in this process	Status
YES Downstream Asher	Descum / wafer clean before deposition and coating	
SPTS PECVD (PECVD2)	Deposits the SiO ₂ hard mask	
Filmetrics F50	Thin-film reflectometer — still useful going forward for bulk oxide/resist thickness measurement	
Rite Track Coater / Developer	Spin-coats and develops AZ10XT photoresist	
ABM Semi-Auto Mask Aligner	Exposes the resist through the current physical photomask	Likely being replaced by MLA150 (below)
MLA150 (Maskless Aligner)	Direct-write lithography — no physical photomask needed. Front-runner for the new mask design, since it avoids the cost/lead-time of writing a new physical mask and gives more design flexibility. Jonah and Labib (other URAs) are running a related MLA characterization project (without oxide) and are expected to help with the oxide-adapted version of this once free.	LEANING DIRECTION, NOT FINALIZED
Nikon Eclipse L200ND	Still useful going forward for optical inspection of develop/lithography quality	
Bruker DektakXT profilometer	Used throughout batch 1/3 for thickness and depth measurement	LIKELY DEPRIORITIZED GOING FORWARD — SEE NOTE BELOW
Nanospec	Used for remaining-resist and oxide thickness measurement	POSSIBLY STILL USEFUL FOR REMAINING-RESIST THICKNESS SPECIFICALLY — TBD
Drop gauge		

	Quick manual thickness check at 5 wafer positions (C/N/E/S/W)	LIKELY DEPRIORITIZED GOING FORWARD — NOT PRECISE ENOUGH
Oxford ICP Fluorine Etcher (ICP-F)	Previously used to etch the SiO ₂ hard mask open	CURRENTLY DOWN. SEE ICP07 BELOW.
ICP07	Replacement oxide-etch tool while ICP-F is down. Grace (WNF etch engineer) has written a new recipe, QT_SiO2_CHF02_600WL1 , for this tool. It still needs to be tested across several wafers before it's locked in as the standard oxide-etch step.	RECIPE EXISTS, VALIDATION IN PROGRESS
Keyence	Primary metrology tool going forward, especially for measuring the oxide-etch stage	
SPTS Silicon DRIE	The Bosch-process deep silicon etch tool — the heart of this project. A vacuum pump issue has been affecting it; Mick's proposed next recipe (higher passivation pressure, lower depassivation/etch pressure) is waiting on Grace's confirmation that the pump is fixed before it's allowed to run.	PUMP STATUS PENDING CONFIRMATION FROM GRACE
Verios FEI SEM	Primary metrology tool going forward for the DRIE stage — for most DRIE verification, this and Keyence should now be the main tools used rather than Dektak/Nanospec/drop gauge.	
EKC 830 Strip Bath	Strips photoresist; also used historically to test whether "cloudy" post-etch wafers clean up	
Class One Avenger Ultra-Pure (SRD)	Wafer rinse/dry	
Vision 320 RIE Mark II	Available; not central to the current recipe	

On metrology going forward, direct from Mick: "we should primarily use Keyence/SEM. Filmetrics is still useful for bulk oxide/resist thickness. Nikon also good for inspecting lithography... should have no need for

Dektak/Nanospec/drop gauge (maybe Nanospec helpful for resist remaining? TBD)." This is a recommendation for how the *process flow going forward* should read — the existing spreadsheet itself has not been edited to match, and doesn't need to be as a prerequisite for this change.

4. Process flow (start to finish)

This is the repeatable fabrication sequence used historically to prepare and etch each test wafer (batches 1 and 3). Full step-by-step detail (recipe names, exact times) as actually run lives in the `Process flow` and `New process flow` tabs of `high_aspect_ratio_etching_recipe_development.xlsx`. Two things have changed since this flow was last run and are called out inline below: the oxide-etch tool/recipe, and the recommended metrology tools.

1. **Ash / descum** (YES Asher, Recipe 4) — clean the bare wafer immediately before oxide deposition.
2. **Deposit SiO₂ hard mask** (PECVD2, target 4 μm, recipe `USER_SiO2_350_4u`) — roughly 9–32 min depending on target thickness and whether the backside is also coated.
3. **Measure oxide thickness** — historically drop gauge / Filmetrics at 5 points (Center, North, East, South, West); **going forward, prefer Filmetrics** for this bulk measurement.
4. **Ash / descum** again immediately before resist coating, to improve adhesion.
5. **Coat photoresist** (Rite Track coater, AZ10XT, 7 μm for the 4 μm oxide target) plus an edge-bead removal (EBR) step so the oxide etch isn't fouled by edge resist buildup.
6. **Rehydrate 30 min. Do not skip** — otherwise the resist cracks after develop.
7. **Measure resist thickness** — prefer Filmetrics.
8. **Expose** — historically the ABM aligner with the HAR DRIE / Uniformity physical mask (84 s total exposure, split into two 42 s intervals with ≥30 s between them). **Going forward, this step is likely to move to the MLA150 maskless aligner** once the new mask design and oxide-adapted process are worked out with Jonah/Labib.
9. **Develop** (Rite Track developer, recipe DEV6).
10. **Inspect** — Nikon optical microscope, still recommended going forward. Confirms clean develop; features under ~3 μm may not resolve visually.
11. **Descum** (YES Asher) — removes residual undeveloped resist via low-power O₂ plasma.

12. **[Checkpoint]** Historically a Dektak scan plus occasional sacrificial cross-section SEM, to verify the resist sidewall profile before committing to the oxide etch. **Going forward, prefer Keyence + SEM here instead of Dektak.**
13. **Etch oxide hard mask.** Historically the Oxford ICP-F, recipe `QT SiO2 Etch-C4F8/O2`, ~20°C, ~5 mTorr, ~10.5 min total, split into 2 sub-etches with a 1-minute rest partway through to reduce resist reticulation. **ICP-F is currently down.** The near-term plan is **ICP07** with Grace's new recipe `QT_SiO2_CHF02_600WL1`, which still needs multi-wafer validation, including confirming it resolves the reticulation/cloudiness issue via better cooling (see Section 10).
14. **Measure remaining resist and etched oxide** — historically Nanospec/Dektak; **going forward, prefer Keyence**, especially for this oxide-etch stage. Nanospec may still be useful specifically for remaining-resist thickness — unconfirmed, treat as TBD.
15. **[Checkpoint]** Sacrificial cross-section SEM — confirm the oxide is fully cleared in the trenches and the sidewall still looks good.
16. **DRIE Bosch etch** (SPTS Silicon DRIE) — the main high-aspect-ratio etch; see Section 5 for the recipe itself. Target trench depth for a full run has been ~500 µm, though individual wafers have been pushed deeper by re-running loop sets on an already-etched wafer (see Section 7).
17. **Measure remaining oxide mask and trench/hole depth.** Historically Dektak/Nanospec/Keyence depending on depth; **going forward, this stage should mainly rely on SEM** (with Keyence as a secondary check), not Dektak/Nanospec/drop gauge.
18. **Final SEM** — sacrifice the wafer, cleave across the features, cross-section at a 45° stage tilt, sputter-coat Au/Pd (~30 s) to reduce charging, and image to verify final etch depth, sidewall profile, and measure aspect ratio directly. See Section 8 for important caveats on exactly what "measure aspect ratio" means in practice.

Cleaving tip (from the lab notes): make the diamond scribe mark on the *backside* of the wafer, and approach the trench array from the **5 µm side** rather than the 500 µm side, so cleaving doesn't damage the narrow high-aspect-ratio features you actually want to image.

5. The DRIE recipe & loop structure

The working recipe family is named `MM_Deep_*` (inherited from prior WNF process work). Recipes with the `MD_` prefix (Mick Deines' initials) are Mick's proposed/trial variants layered on top of that base recipe. The full, tool-level recipe (every process step's pressure, RF power, gas flows, capacitor tuning, etc.) is stored in `SPTS_DRIE_recipe.xlsx`, tabs `MM_Deep_3x_-3` and `standard Si etch` — that file is the actual tool-importable recipe definition, not just a summary, so treat it as the source of truth for exact parameter values, and look there for the distinction between the existing baseline recipe and Mick's proposed changes.

Loop structure

Each Bosch "loop" is a **passivation** → **depassivation** → **etch** trio (see Section 2). A full etch consists of many such loops run back to back.

Two recipe variants

- `MM_Deep_STD_-3` ("standard," also referred to as `MM_Deep_Standard_-3` in some logs — same recipe) — a single loop-set, repeated for the whole etch. Simpler, lower selectivity.
- `MM_Deep_3x_-3` ("3x") — three sequential loop sets (Set 1 → Set 2 → Set 3), each with its own passivation/depassivation/etch timing and power, run one after another. Splitting the etch into three parameter regimes lets the recipe compensate for how ion/neutral transport into the trench changes as it gets deeper. This is the recipe family this project is actively tuning.

What differs between the three loop sets

Parameter	Set 1 (shallow)	Set 2 (mid)	Set 3 (deep)
Deposition (passivation) time	1.2 s	1.2 → 1.4 s	1.4 s
Etch time	1.2 s	1.2 → 1.5 s	1.5 s
Etch HF (platen bias) power	64 → 70 W	70 → 80 W	80 → 60 W

(Source: `DOE` tab, `high_aspect_ratio_etching_recipe_development.xlsx`.) The **third loop set** is the **main tuning target** — the original next step was explicitly to modify the third loop set of the 3x recipe

to increase etch rate/aspect ratio, and that's what the 8-trial JMP DOE (Section 6) varies. Mick's proposed next recipe change (pending Grace's pump confirmation, see Section 11) is to **increase passivation pressure and decrease depassivation/etch pressure**, to improve etch directionality.

6. The Design of Experiments (DOE)

JMP statistical software was used to choose 2–3 parameters to vary (with upper/lower bounds) for the deep/third loop set, producing an 8-trial fractional-factorial-style experiment. The four parameters varied were passivation pressure, depassivation pressure, platen (bias) power, and etch step pressure/time:

Trial	Passivation pressure	Depassivation pressure	Platen power (W)	Etch pressure	Etch time (s)
1	45 mTorr	35 mTorr	80→60	35 mTorr	2.5→3
2	45 mTorr	35 mTorr	80→60	35 mTorr	2.5
3	35 mTorr	25 mTorr	80→100	25 mTorr	2.5
4	45 mTorr	35 mTorr	80→100	35 mTorr	2.5→3
5	35 mTorr	25 mTorr	80→60	25 mTorr	2.5→3
6	35 mTorr	25 mTorr	80→100	25 mTorr	2.5→3
7	45 mTorr	35 mTorr	80→100	35 mTorr	2.5
8	35 mTorr	25 mTorr	80→60	25 mTorr	2.5

Each trial recipe was saved on the tool as MD_DOE1_3x_-3 through MD_DOE8_3x_-3 and run on a wafer from the first batch (wafers A9–A16). Per-wafer run logs (dates, He flow, reflected power, loop counts actually used) are in the SPTS-DRIE tab — **treat that tab as the accurate record of exactly what ran**, since a couple of wafers were etched more than once under different recipe entries (see wafer A9 in Section 7). Measured depths/thicknesses are in DOE data and 5 um depth tabs.

7. Results so far

Early process verification (batch 1, wafers A1–A8)

- **Wafer A1** — confirmed the resist sidewall profile looked correct after lithography, before committing to etching.
- **Wafer A4** — 10.5 minutes in the ICP-F oxide etch; slightly overetched but acceptable, confirming the ICP-F recipe was close to dialed in (this recipe is now moot since ICP-F is down — see Section 3).
- **Wafer A5** (MM_Deep_STD_-3 , 500 loops) — etch rate 0.60 $\mu\text{m}/\text{loop}$, **reported selectivity 80.3**, aspect ratio $\approx 26:1$.
- **Wafer A6** (MM_Deep_3x_-3 , 945 loops) — etch rate 0.63 $\mu\text{m}/\text{loop}$, **reported selectivity 139.3**, aspect ratio $\approx 47:1$ — the best clean result to date, roughly double the previous WNF record and already close to the low end of the 50–100:1 target range.

Why A5's and A6's selectivity numbers look inconsistent with the DRIE etch rates tab

Selectivity ($\text{Si etch rate} \div \text{oxide erosion rate}$) is **trench-width dependent**: a narrower trench etches silicon more slowly (ARDE, Section 2) while the top oxide mask erodes at roughly the same rate regardless of trench width, so a narrower trench shows up as lower selectivity. The DRIE etch rates tab appears to mostly reflect the wide (500 μm) trench, which etches faster and so reports a higher selectivity than the 80.3 figure quoted for A5 in the slide deck — that 80.3 is probably based on a narrower, slower-etching trench. Neither number is "wrong," but the spreadsheet doesn't currently document which trench width each selectivity figure corresponds to, which is exactly the kind of ambiguity worth fixing (see the callout in Section 1).

Wafers A7/A7b/A7c/A8 continued pushing the 3x recipe to deeper total depths (up to $\sim 800\text{--}850\ \mu\text{m}$) by re-running just the third loop set on an already-etched wafer, to explore the tool/wafer's practical depth limits. Detailed etch-rate/erosion numbers for every wafer are in the DRIE etch rates tab.

Note on wafer A9: the SPTS-DRIE log shows A9 etched under two different DOE recipe entries. The best explanation: a short, low-loop-count run first (possibly to confirm the new settings wouldn't trip a tool alarm), followed by the full run with loop set 1 reduced to compensate. Not considered a significant issue for interpreting the data, but flagged here since it could otherwise look like a logging error.

DOE trend findings (batch 1, wafers A9–A16, trials 1–8)

- **Etch time** (longer) and **platen power** (lower) each independently improved aspect ratio — this became the headline recommendation.
- **Pressure** changes alone had a *negligible* effect on aspect ratio — a surprise, since lower pressure was expected to reduce off-axis ion scattering and improve aspect ratio (Section 2).
- There was a significant **interaction effect**: simultaneously decreasing passivation pressure *and* increasing platen power caused a large drop in aspect ratio (this shows up as an outlier trial). Excluding that outlier, the pressure-vs-aspect-ratio trend matches the originally expected direction.
- SEM cross-sections of Trial 3 vs. Trial 4 show a visibly large difference in aspect-ratio-dependent etching (bowing/undercut).

Working conclusion: the resulting recommended recipe direction is *high passivation pressure, low etch pressure, low platen power, and long etch time* for the deep (third) loop set. This lines up with Mick's proposed next recipe change (Section 5) — it's currently blocked on Grace confirming the DRIE tool's pump issue is resolved.

8. Aspect ratio: how it's measured, and why the number is squishy

This section exists because "what's our best aspect ratio?" doesn't have a single clean answer — and pretending otherwise would make this document less useful, not more.

Best reported figure: ~60:1, from roughly 300 μm of etch depth in a nominally 5 μm -wide trench. That number should be treated as an **optimistic upper-bound estimate**, not a validated measurement, for two independent reasons:

1. **The depth side is mostly solid, but check which depth number is used.** Most of these figures are pulled from the `5 um depth` tab, which computes several depth estimates per wafer from different metrology sources (drop gauge, Keyence, SEM, all scaled against each other). The **"Pseudo-depth" column is the correct one to use** — the other columns in that tab reflect a fairly flawed measurement methodology and shouldn't be used directly for aspect ratio calculations.

2. **The width side is the bigger problem.** Aspect ratio calculations generally assume the trench stayed at its nominal 5 μm mask-opening width for the entire etch depth. In practice this isn't true — surface/sidewall damage near the top of the trench can widen the effective opening to nearly 20 μm for some recipes. An aspect ratio computed as (pseudo-depth $\div$ nominal 5 μm) is therefore an overestimate whenever that widening happened; the true local aspect ratio deeper in the trench, or an aspect ratio computed against the actual (damaged) top width, would be meaningfully lower.

Extra caution for wafers A9, A14, and A15: for these three wafers specifically, the pseudo-depth calculation had to fall back on drop-gauge-based scaling because the oxide mask broke through (was fully consumed) during those etches — meaning the mask stopped protecting the surrounding surface partway through, adding another layer of estimation uncertainty on top of the general caveats above. Treat any aspect ratio figures for A9/A14/A15 as the least reliable in the dataset.

Suggested task for the new students OPEN

Nail down and document one consistent, defensible aspect-ratio definition (e.g., pseudo-depth $\div$ measured actual top width from SEM, per wafer, with the definition written down somewhere visible in the workbook), then recompute the headline numbers with it. This will very likely change the reported $\sim 47:1$ (A6) and $\sim 60:1$ figures, hopefully making them more defensible rather than necessarily higher. Cross-referencing the SEM archive (Section 9) — measuring actual top-of-trench width directly off annotated images — is the way to get real data for the width side of this calculation.

9. SEM image archive (GitHub)

Raw and annotated SEM cross-section images are **not embedded in this document** — they live in a companion GitHub repository, which an AI assistant should fetch and search directly when a question requires looking at actual imagery (e.g., "what did wafer A14's sidewall actually look like," or the feature-size investigation in Section 3).

Repository: github.com/mickdeines/HAR-DRIE

The repo root also contains copies of the other project files listed in Section 14. The SEM images specifically are inside three ZIP archives — `1mm Si Part 1.zip`, `Part 2.zip`, `Part 3.zip` — the 3-way split is arbitrary (just file size), not organizational. There is no README or manifest in the repo, so this section is the index.

Folder / filename convention

```
1mm Si/<Wafer>/<Position>/<Raw|Annotated>/  
<Wafer>_[<Position>_]<##>[_e].jpg
```

- **Wafer** — e.g. `A6`, `A14`. Covers batch-1 wafers A1, A2, A4–A16 (no A3). **No batch-3 (B-series) images exist yet** — consistent with batch 3 not having reached the DRIE/SEM stage (Section 11).
- **Position** — `C` (center), `S` (south), or `Unspecified` (position on the wafer wasn't recorded for that image). Other mask positions (N, E, W) exist in the process flow's measurement convention but aren't represented in this particular image set.
- **Raw vs. Annotated** (confirmed by inspection): **Raw** images are the plain SEM capture, tightly cropped, with no on-screen overlay. **Annotated** images (filename ends `_e`) add back the instrument's HUD bar (facility name, detector, accelerating voltage, magnification, working distance, scale bar) and, for a subset of images, hand-added cyan measurement callouts (depth/width dimensions, sidewall angle) — e.g. `A14_C_01_e.jpg` has eight labeled trench-depth measurements across varying trench widths in one shot. Not every Annotated image has measurement lines — some just add the HUD bar back. There's no separate flag distinguishing which is which; you have to open the image.

How an AI assistant should use this: clone or download the ZIPs, extract by wafer ID (e.g., `unzip -l "1mm Si Part 1.zip" | grep A14` to find what's available), and prefer **Annotated** images when the question needs a measurement or scale reference, or **Raw** when a cleaner, unobstructed view is more useful. There are 207 images total, unevenly distributed across wafers (A9 has 28; A1 has only 6).

10. Practical lessons & status of open issues

- **Cloudy wafers after the oxide etch** **ROOT CAUSE IDENTIFIED** — root cause is now understood to be **resist reticulation from the wafer overheating** during the oxide etch. Fix: better cooling during that step. This should be tested and confirmed as part of validating the new ICP07 recipe (`QT_SiO2_CHF02_600WL1` , Section 3) across multiple wafers.
- **New mask decision** **LEANING, NOT FINAL** — leaning toward **maskless alignment (MLA150 tool)** for the new small-feature design, rather than writing a new physical photomask. This gives easier lithography and more design flexibility, but depends on Jonah and Labib (other URAs) finishing their related non-oxide MLA characterization project and then helping adapt it for a design that includes oxide.
- **Backside protection** **RESOLVED** — not expected to be needed once the mask moves to narrower/smaller trenches, since those etches won't need to go nearly as deep.
- **Oxide thickness (6 μm plan)** **DEPRIORITIZED** — shelved in favor of reverting to 4 μm oxide / 7 μm resist; not worth the added lithography difficulty right now. Final call left to Caitlynn/Samyama.
- **ICP-F down** **OPEN — IN TESTING** — switched to ICP07 with a new recipe from Grace; needs validation across several wafers before it's the standard.
- **DRIE pump issue** **OPEN — WAITING ON GRACE** — Mick's proposed next DRIE recipe change (higher passivation pressure, lower depassivation/etch pressure) is waiting on Grace confirming the pump is fixed.
- **Maximum safe feature size for the new mask** **OPEN — SEE SECTION 3** — needs investigation using the SEM archive before the new mask design is finalized, due to pillar-collapse/breakthrough risk.
- **Aspect ratio definition** **OPEN — SEE SECTION 8** — needs to be standardized and documented in the workbook.
- **General data-taking recommendation** (still good advice): take data live during etches rather than only reviewing it afterward, or set up live graphs of key parameters — this makes it much easier to catch trends and problems as they happen, especially on long DRIE runs.

11. Current status & next steps

Batch 3 (wafers B01–B12) has not progressed past the early oxide/lithography steps — no DRIE or SEM data exists for it yet. The project is effectively paused at a decision point on wafer/mask direction while several dependencies resolve. In rough priority order:

1. **Decide (or confirm) the oxide/resist revert.** Mick's recommendation is 4 μm oxide / 7 μm resist, abandoning the thicker 6 μm oxide / 9–10 μm resist plan — final call is Caitlynn/Samyama's to make.
2. **Validate the ICP07 oxide-etch recipe** (`QT_SiO2_CHF02_600WL1`) across several wafers, confirming it also resolves the reticulation/cooling issue.
3. **Reach out to Jonah and Labib** to specify what an oxide-adapted version of their MLA150 characterization work would need to look like, once they're free.
4. **Wait for Grace's confirmation** that the SPTS DRIE pump issue is resolved before running Mick's proposed passivation/depassivation-pressure recipe change.
5. **Plan the next DOE round** around the new mask (smaller features) and 500 μm -thick wafers, once the mask/lithography path is settled — including the max-feature-size investigation from Section 3.
6. **Clean up the data workbook:** document a consistent aspect-ratio definition (Section 8), and clarify which trench width each selectivity/etch-rate figure corresponds to (Section 7).
7. **Continue toward the 50:1–100:1 target** aspect ratio range. Best reported result to date is $\sim 60:1$, but see Section 8 for why that number needs re-verification before being treated as solid.

Reaching Mick: Mick is available via Microsoft Teams through September 15, 2026, for questions on anything unclear in this document or the underlying data. After that, Grace Tong is the primary WNF-side contact.

12. People, roles & funding

- **Mick Deines** (Chemical Engineering, UW) — led the project through the work described here; reachable via Teams through Sept 15, 2026.

- **Caitlynn** — incoming student, currently leading planning (mask/process decisions, coordinating with Jonah/Labib and Grace).
- **Samyama** — incoming student, will take on more hands-on fabrication/etching work once the plan is settled.
- **Qi "Grace" Tong** (WNF staff) — Grace is the etch engineer providing the new ICP07 oxide recipe and confirming DRIE pump status; she has mentored the process development throughout.
- **Karl Böhringer** (ECE, UW) — faculty mentor / co-author.
- **Jonah and Labib** — other undergraduate research assistants, currently finishing a related maskless-alignment (MLA150) HAR characterization project without oxide. Expected to help adapt their work for an oxide-based design once free — this project needs to define those requirements for them.
- **Cameron Toskey** and **Jean Nielsen** — helped with DOE setup and lithography steps historically.
- **Darick Baker** — manages the WNF undergraduate program.
- **Funding/acknowledgment:** supported by the Microelectronics Commons Program (a Department of War initiative), and conducted at the Washington Nanofabrication Facility / Molecular Analysis Facility, an NNCI site at UW, with partial NSF support (awards NNCI-1542101 and NNCI-2025489).

13. Reference literature

Two outside papers were kept alongside the project files as background/benchmarking references (the "Important Resources" notes also name a third ramped-parameter-process paper not included in this file set):

Marty et al., "Very High Aspect Ratio Deep Reactive Ion Etching of Sub-micrometer Trenches in Silicon" (2012)

Reports Bosch-process aspect ratios up to about 160:1 on 250 nm-wide trenches, and about 120:1 on 35 nm trenches using a cryogenic (rather than Bosch) process, for sub-micrometer-scale features — a different feature-size regime than this project's 5–500 μm trenches, so results aren't directly comparable, but the paper's proposed empirical model (aspect ratio as a logarithmic function of trench width, fit per etch process) is a useful framework for

predicting how a given recipe's aspect ratio should scale with feature size — potentially relevant to the feature-size investigation in Section 3.

Kim & Lee, "Optimization of deep reactive ion etching for microscale silicon hole arrays with high aspect ratio" (Micro and Nano Systems Letters, 2022)

Focuses on etching high-aspect-ratio *hole* arrays (harder than trenches, because ion/radical transport is restricted in all directions rather than just two). The authors optimize bias voltage and ICP source power to promote etching at the bottom of the hole, and separately tune C_4F_8/SF_6 flow rates to reduce sidewall undercut, reaching an aspect ratio of about 20:1 on 1.3 μm -diameter holes. Relevant if this project extends from trenches to hole arrays, since the proposal explicitly targets both geometries.

14. File inventory

github.com/mickdeines/HAR-DRIE

Companion repository holding the same project files as below, **plus the SEM image ZIPs** described in Section 9. Check here first for anything requiring actual imagery.

HAR_DRIE_Project_Proposal.pdf

The original project proposal: goals, target aspect ratio (50–100:1), suggested approach, tool-training checklist, resources, and a week-by-week time estimate for the quarter.

DRIE_High_Aspect_Ratio_DOE_slides.pptx

Short internal progress-update slide deck: wafer-by-wafer summary (A1, A4, A5, A6) and the "next steps" that launched the JMP DOE (Section 6).

Mick_Deines_HAR_DRIE_DoE_Intern_Showcase.pptx

The end-of-internship presentation: background on RIE/Bosch/aspect ratio, methods, the 8-trial DOE table, results trends, SEM comparison, conclusions, and acknowledgments. Good single overview of the whole project as of that point.

Development_of_High_Aspect_Ratio_Deep_Reactive_Ion_Etching_Using_Design_of_Experiments

(and duplicate

..._1.pdf

) The research poster version of the same project — most complete single-page summary of introduction, process flow, DOE table, results plot, and conclusion.

Bosch_Etching_Title_Page.docx

Title/cover page for a more formal final report (mostly placeholder fields) — states the 50:1–100:1 target explicitly. The full report body doesn't appear to have been written yet.

2Very_High_aspect_Ratio_Deep_Reactive_Ion_Etching_of_Sub-micrometer_Trenches_in
and

3ImprovedHAR_hole_Bosch_2022.pdf

Reference journal papers — see Section 13.

SPTS_DRIE_recipe.xlsx

The actual tool-level recipe definitions. Tab `MM_Deep_3x_3` : full step-by-step parameters for the 3-loop-set recipe, including the baseline and Mick's proposed (`MD_` - prefixed) variants. Tab `standard Si etch` : the simpler single-loop-set recipe. Treat this file as ground truth for exact recipe values.

high_aspect_ratio_etching_recipe_development.xlsx

The main lab notebook / data workbook — **not yet cleaned up; see Sections 1, 7, and 8 for known ambiguities**. Key tabs:

- `Process flow` / `New process flow` — the fabrication sequence (Section 4), original and revised.
- `DOE` — the loop-set-by-loop-set parameter table (Section 5) and the DOE parameters (Section 6).
- `DOE data` — measured results for wafers A7–A16, including the DOE trial wafers.
- `SPTS-DRIE` — per-run tool log: date, recipe name used, loop counts per set, He flow, reflected power. **Authoritative record of what actually ran**, including the A9 double-run (Section 7).
- `ICP-F` — per-run oxide-etch log for the (now-down) ICP-F tool.
- `DRIE etch rates` — computed etch rate/erosion/selectivity per wafer; see Section 7 for why these numbers look inconsistent across sources.

- **5 um depth** — detailed 5 μm -trench depth measurements used to compute reported aspect ratios; **use the "Pseudo-depth" column** (Section 8), not the other depth columns.
- **batch 1 data** / **batch 3 data** — full per-wafer travelers (every process-flow measurement, step by step). Batch 1 = wafers A1–A16. Batch 3 = wafers B01–B12 (stalled at early steps — no batch 2 tab exists in this workbook).
- **Drop gauge** — raw drop-gauge thickness readings (historical; deprioritized going forward, Section 3).
- **Important Resources** — free-text lab meeting notes, largely superseded by the clarifications in this document (Sections 10–11), plus links to the reference papers and slide deck.
- **MM_Deep_3x_-3** — a copy of the tool recipe (same content as the dedicated recipe file above).

15. Suggested first steps

1. Read the intern showcase poster/slides for the fastest full-project overview, then this document for what's changed and what's still open since then.
2. Get trained/qualified on the WNF tool list in Section 3 if not already, **noting the ICP-F → ICP07 and Dektak/Nanospec/drop-gauge → Keyence/SEM shifts** before assuming the original proposal's tool list is current.
3. Message Mick on Teams (through Sept 15) with any ambiguity that comes up early — after that, Grace is the contact.
4. Reach out to Jonah and Labib to start scoping the oxide-adapted MLA150 requirements — this is on the critical path for the new mask direction.
5. Check in with Grace on both open tool questions: the ICP07 recipe validation, and the DRIE pump status blocking Mick's proposed recipe.
6. Before finalizing new-mask feature sizes, do the SEM/data investigation into safe minimum feature size (Section 3) — this affects the mask design Jonah/Labib will need to help build.
7. As an early, genuinely useful task: work through Sections 7–8 of this document against the actual workbook, and start documenting a consistent aspect-ratio definition and trench-width-labeled selectivity/etch-rate figures directly in the spreadsheet.

Passdown document compiled from project files and a series of clarifying conversations with the outgoing student (Mick Deines), for handoff to incoming WNF undergraduate researchers Caitlynn and Samyama. Source materials: project proposal, progress slides, final poster/report draft, tool recipe files, lab data workbook, and the companion GitHub SEM archive, as of September 2026.